

# Histidine-Rich Glycoprotein: A Possible Modulator of Coagulation and Fibrinolysis

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## ABSTRACT

Histidine-rich glycoprotein (HRG) is one of the major plasma proteins and thought to function in blood coagulation, fibrinolysis, and innate immune systems. The amino acid sequence of HRG revealed a multidomain structure consisting of cystatin-like domains 1 and 2, a Pro-rich domain 1, a His-rich domain, a Pro-rich domain 2, and a C-terminal domain. Broad ligand-binding properties of HRG are involved in the multivalent functions of HRG. Among various functions of HRG, its interactions with heparin/heparan sulfate, fibrinogen, and plasminogen are thought to be intimately related to its roles in the coagulation and fibrinolytic systems. Recent studies on these topics are mainly reviewed in this article. The newly disclosed abilities of HRG in angiogenesis, its antibacterial effect, its activation of T-cell lines in cooperation with Concanavalin A, and the identification of a putative receptor for HRG on T cell lines are also described.

**KEYWORDS:** Histidine-rich glycoprotein, HRG, coagulation, fibrinolysis

Almost 40 years has passed since the discovery of histidine-rich glycoprotein (HRG), but its precise physiological function still remains enigmatic. HRG was first isolated as one of four serum proteins adsorbed tightly to carboxymethyl-cellulose.<sup>1</sup> It has a very high content of histidine and thus was named “histidine-rich 3.8S- $\alpha_2$ -glycoprotein.” From the late 1970s to the 1980s, many properties of HRG were reported. The interactions of HRG with various molecules such as divalent metal ions,<sup>2</sup> plasminogen,<sup>3</sup> fibrinogen,<sup>4</sup> and glycosaminoglycans such as heparin or heparan sulfate<sup>5,6</sup> were reported, and HRG was thought to function as a modulator molecule of blood coagulation and fibrinolysis. Elevated plasma levels of HRG in patients with thrombotic disorders had been reported, and high plasma HRG levels were once assumed to be a risk factor for throm-

bosis.<sup>7</sup> However, subsequent studies could not establish a causal relationship between the elevated plasma level of HRG and thrombophilia.<sup>8</sup> Paradoxically, HRG deficiencies in thrombotic patients were also reported,<sup>9–11</sup> and the function of HRG in the blood coagulation system became obscure. In addition, phenotypic analyses of the HRG-knockout mice could not confirm a tight relationship between plasma HRG and hemostatic abilities.<sup>12</sup> Thereafter, the major research focus on HRG shifted to its possible functions in angiogenesis,<sup>13</sup> antibacterial,<sup>14</sup> and immunological systems. This review summarizes recent progress concerning studies on the function of HRG in hemostasis and fibrinolysis, and topics in the immunological or other fields are briefly described. Readers interested in these fields are advised to consult with recent reviews.<sup>15,16</sup>

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Forgotten Factors in Thrombosis and Hemostasis; Guest Editors,

Marcel Levi, M.D., Ph.D., and Joost C.M. Meijers, Ph.D.

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## STRUCTURE OF HISTIDINE-RICH GLYCOPROTEIN

The amino acid sequence of human HRG was deduced from the nucleotide sequence of its cDNA.<sup>17</sup> The total number of amino acids in mature human HRG is 507 and the number of His residues is 66 (13%). It has five potential carbohydrate attachment sites including one Asn-Asp-Cys sequence at position 69. The most remarkable feature of human HRG is its internal repeat structure, that is, two cystatin-like domains at the N-terminal region, a His-rich domain that is flanked by two Pro-rich domains, and a C-terminal domain. The His-rich domain that contains about a half of the His residues in HRG (32 of 66) is arranged in 12 repeats of a consensus Gly-His-His-Pro-His sequence and induces a conformational change in the whole molecule upon chelating to metal ions, especially to  $Zn^{2+}$ . The importance of this conformational change is discussed later. The N-terminal 229 residues constitute two cystatin-like domains, and consequently, HRG is classified in the cystatin superfamily.<sup>18</sup> Besides human HRG, HRGs have been isolated from plasmas of several mammalian species including bovine, mouse, rat, and rabbit. Each of these HRG preparations are composed of ~500 amino acid residues and exhibit a molecular weight of ~70,000, which includes one or more carbohydrate moieties. Interestingly, two molecular species of HRG were obtained from rat plasma.<sup>19</sup> The gross structures included five intramolecular disulfide bonds (two within cystatin domain 1, one in cystatin domain 2, one between cystatin domain 1 and C-terminal domain, and one between cystatin domain 2 and Pro-rich domain 2), which are highly conserved among mammalian HRGs. These disulfide bonds keep the molecule from dispersion upon proteolytic digestion. However, the carbohydrate attachment sites are not well conserved among mammalian HRGs. It is particularly noteworthy that chicken HRG is fairly different from mammalian HRG. It is composed of 721 amino acid residues and its molecular weight is ~100,000, including N- and O-linked carbohydrate chains. It also has two cystatin-like and C-terminal domains homologous to mammalian HRGs, but the central portion is composed of three His/Pro-rich regions and two Pro-rich regions instead of two Pro-rich regions and one His-rich region found in mammalian HRG.<sup>20</sup>

The human HRG gene is ~15.5 kb long and consists of 7 exons and 6 introns.<sup>21</sup> Each of the cystatin-like domains is encoded by three exons similar to those of other members of the cystatin superfamily; the entire C-terminal half of the HRG molecule is encoded by a single large exon. Following the determination of its complete genome sequence, the precise location of the HRG gene was assigned on 3q27,<sup>22</sup> proximal to the genes for other members of the cystatin superfamily such as  $\alpha_2$ -HS glycoprotein, fetuin B, and kininogen. In the

rat genome, two HRG genes are closely located on 11q23. The HRG gene is probably derived from the ancestral cystatin gene by multiple gene duplications in this region as predicted from these genomic structures.

## FUNCTIONS IN COAGULATION AND HEMOSTASIS

### Changes in the Plasma Level Related to Physiological and Pathological States

Human HRG is present in plasma at relatively high concentrations (100 to 150 mg/L).<sup>1,23,24</sup> HRG mRNA was detected in cultured rat, monkey, and human hepatocytes, and further studies on the production of mRNA revealed that the liver is the sole source of plasma HRG,<sup>25</sup> although HRG is also found in platelets, megakaryocytes,<sup>26</sup> colostrum, and milk.<sup>27</sup> HRG is thought to be taken up from plasma because there is no positive evidence that cells other than liver parenchymal cells produce mRNA for HRG. The plasma HRG level is low in the neonatal stage and gradually increases with age.<sup>23</sup> Serum HRG decreases during pregnancy and with a low dose of oral contraceptives containing levonorgestrel or lynestrenol,<sup>28</sup> and it is assumed that estrogens are responsible for these changes. The HRG level in plasma is reduced in either advanced liver cirrhosis,<sup>29</sup> sepsis, or in the acute phase.<sup>30</sup>

### Knockout Mouse

To elucidate the physiological function of HRG, HRG-knockout mice were produced and their hemostatic factors analyzed.<sup>12</sup> These mice were viable and fertile. There were no appreciable abnormalities in their hemostatic components except for the absence of HRG in their blood. The fibrinogen levels in female  $Hrg^{-/-}$  mice were somewhat higher (135%), and the number of monocytes in  $Hrg^{-/-}$  mice were significantly higher (252%) than those of their wild-type littermates. Also, some differences in blood tests were noted. The prothrombin time was shorter in  $Hrg^{-/-}$  mice than in  $Hrg^{+/-}$  and  $Hrg^{+/+}$  mice. Bleeding time after tail tip amputation was shorter in  $Hrg^{-/-}$  mice than in  $Hrg^{+/+}$  mice, and the spontaneous fibrinolytic activity in clotted blood was higher in  $Hrg^{-/-}$  mice than in  $Hrg^{+/+}$  mice. These differences may not pose problems in the life of a healthy HRG-deficient mouse but may affect the fate of mice under some abnormal or pathological condition.

### Interaction with Heparin and Heparan Sulfate

Since the first report on a high degree of sequence homology between the N-terminal regions of HRG and antithrombin III (currently, antithrombin [AT]) where the putative heparin-binding site of AT is located,<sup>5</sup>

the interaction between heparin and HRG has been well studied. HRG binds heparin and thereby interferes with its interaction with AT, resulting in the neutralization of its anticoagulant activity. The interaction between heparin and HRG is counteracted by  $\text{Ca}^{2+}$ -binding anticoagulants, indicating that it depends on the presence of divalent metal ions.<sup>6</sup> Both HRG and platelet factor 4 effectively neutralize the ability of heparin to accelerate the inhibitions of both activated protein C and thrombin by protein C inhibitor (PCI). The heparin-neutralizing ability of HRG in the APC inhibition by PCI, however, was decreased in a  $\text{Ca}^{2+}$ -dependent manner and apparently lost at 1 mM  $\text{Ca}^{2+}$ , whereas it was enhanced by  $\text{Zn}^{2+}$ , regardless of the presence or absence of  $\text{Ca}^{2+}$ .<sup>31</sup> The binding site of heparin in HRG was predicted to be the N-terminal region from its homology to AT, and the chymotryptic fragment of HRG containing its N-terminal cystatin domains 1 and 2 has been shown to have heparin-binding and heparin-neutralizing ability comparable with the native HRG molecule.<sup>32</sup> The cell surface heparan sulfate binding site of HRG was analyzed by flowcytometry.<sup>33</sup> HRG was shown to bind to many cell lines in a  $\text{Zn}^{2+}$ -dependent manner but failed to interact with the CHO cell line lacking cell-surface glycosaminoglycans. Heparinase treatment of the normal CHO cells abolished HRG binding. It was also shown that the N-terminal domain of HRG produced by recombinant technology bound specifically to immobilized heparin and cell-surface heparan sulfate in a  $\text{Zn}^{2+}$ -independent manner. In contrast, synthetic peptides corresponding to the  $\text{Zn}^{2+}$ -binding His-rich domain of HRG did not interact with cells. From these results, it was proposed that the N-terminal domain binds to cell-surface heparan sulfate and the interaction of  $\text{Zn}^{2+}$  with the His-rich domain can indirectly enhance cell-surface binding.<sup>33</sup> However, Vanwildemeersch et al showed a 35-amino-acid residue synthetic peptide, HRG330–364, derived from the His-rich domain, retained the  $\text{Zn}^{2+}$ -dependent inhibitory effect on blood vessel formation in vitro and in vivo.<sup>34</sup> HRG330–364 bound heparin/heparan sulfate with the same capacity as full-length HRG and the binding was  $\text{Zn}^{2+}$  dependent. Peptides derived from the His-rich domain of HRG downstream of HRG330–364 failed to inhibit endothelial cell migration and displayed a significantly reduced heparin-binding capacity. An even shorter peptide, HRG335–360, covering a 26-amino-acid sequence within HRG330–364, retained full heparin/heparan sulfate-binding capacity.<sup>34</sup> Therefore, the real interaction site in HRG to heparan sulfate remains controversial.

### Interaction with Fibrinogen and Fibrin

The interaction of HRG with fibrinogen (Fbg)/fibrin was first reported in 1986 by Leung.<sup>4</sup> HRG specifically bound to Fbg immobilized to a plastic plate in a dose-

dependent manner, and the dissociation constant was 6.7 nM. The binding was abolished in the presence of EDTA. The proteolytic fragment of human HRG, 361–389, consisting of the putative repetitive sequence of GHHPH, was shown to specifically bind to Fbg in a  $\text{Zn}^{2+}$ -dependent manner by Interaction Analysis System (IASys). A synthetic His-rich peptide (HHPHG)<sub>4</sub> also bound to Fbg in a  $\text{Zn}^{2+}$ -dependent manner, demonstrating that the His-rich domain is responsible for the binding of HRG to Fbg.<sup>35</sup> The HRG-binding site in Fbg was also analyzed. Fbg2, an isoform of Fbg, which is the product of the alternatively spliced form of the Fbg  $\gamma$ -chain mRNA, has an extra C-terminal region rich in acidic amino acid residues and provides the thrombin and factor XIII binding site. HRG binds to both forms of Fbg, normal Fbg1 and this alternative form Fbg2, but more efficiently to Fbg2. The His-rich domain of HRG binds to this C-terminal acidic region of Fbg2 in the presence of  $\text{Zn}^{2+}$ . HRG also binds to Fbg1 and, therefore, some portion of Fbg other than  $\gamma$ -region is also responsible for HRG binding.<sup>36</sup> Tryptic digestion of Fbg revealed that the major binding site is located in the D-region, and further fragmentation revealed that this disulfide-bonded fragment effectively inhibited the interaction between Fbg and HRG. This fragment was derived from the interchain ring C, at the C-terminal end of the coiled-coil region, and the cleavage of disulfide bonds by performic acid abolished its inhibitory activity (Wakabayashi et al, unpublished results). HRG was reported to prolong the thrombin time.<sup>4</sup> Because HRG and/or Zn did not affect the release of fibrinopeptides A and B by thrombin, the prolongation effect of HRG on thrombin time is due to the inhibition of assembly of fibrin monomer.<sup>37</sup> These findings, together with the shorter bleeding time in HRG-deficient mice, strongly suggest an anticoagulant effect of HRG.

### Interaction with Thrombospondin

HRG is also found in  $\alpha$ -granules of platelets and released by thrombin stimulation.<sup>26</sup> Human HRG forms a complex with purified human platelet thrombospondin. The formation of the thrombospondin-HRG complex is specific, concentration-dependent, and saturable.<sup>38</sup> HRG binds to activated platelets and remains surface bound, and thrombospondin is a likely candidate for a HRG receptor on the platelet surface.<sup>39</sup> This surface-bound HRG can form a complex with heparin or plasminogen, as described later, and modulates coagulation or fibrinolysis, respectively. Activated platelets aggregate through a complex formation between an activated surface glycoprotein (GPIIb/IIIa) and Fbg. As described earlier, HRG can bind to Fbg and therefore modulate the platelet aggregation process. Preliminary data suggest that HRG can interfere with the interaction between GPIIb/IIIa and Fbg in a

Zn<sup>2+</sup>-dependent manner although the extent of regulation needs to be explored further (Kusuma et al, unpublished results).

HRG also contains a CD36 homology domain and acts as a soluble decoy to block the antiangiogenic activity of thrombospondin, thereby promoting angiogenesis. The tripartite interactions among CD36, thrombospondin type I repeat domains, and HRG in tissues thus may play an important role in regulating physiological and pathological angiogenesis.<sup>40</sup>

## FUNCTION IN FIBRINOLYSIS

### Interaction with Plasminogen

HRG was isolated together with  $\alpha_2$ -antiplasmin as an unknown protein bound to an immobilized plasminogen fragment containing the lysine-binding site.<sup>3</sup> Chemical analyses identified this unknown protein as HRG. Because HRG bound to plasminogen with an apparent dissociation constant of 1.1  $\mu$ M, ~50% of the circulating plasminogen in blood is reversibly complexed with HRG. Furthermore, HRG retards fibrinolysis by forming a ternary complex with tissue plasminogen activator (tPA) and plasminogen in the presence of fibrin through interaction with the lysine-binding site of plasminogen.<sup>3</sup> This antifibrinolytic effect of HRG is due to a reduction of the effective plasminogen concentration in blood. On the other hand, by forming a ternary complex with plasminogen and tPA, HRG enhances plasminogen activation 30-fold. This is attributed to the profibrinolytic activity of HRG, and thus activated plasmin is not inhibited by  $\alpha_2$ -antiplasmin.<sup>41</sup> HRG has been shown to function as an adaptor molecule to tether plasminogen to GAG (glycosaminoglycan)-bearing surfaces and to regulate plasminogen activation via various mechanisms.

The HRG-plasminogen interaction is lysine dissociable and involves predominantly the C-terminal heptapeptide sequence of HRG, including the C-terminal lysine, and the Kringle domain of plasminogen.<sup>42</sup>

HRG was also shown to tether plasminogen to cell surfaces, with this interaction potentiated by an elevated level of Zn<sup>2+</sup>, low pH, and conditions that occur at sites of tissue injury, tumor growth, and angiogenesis. Based on these observations, HRG was proposed to act as a soluble adaptor molecule that binds to cells at sites of tissue injury, tumor growth, and angiogenesis, providing a high affinity receptor for tethering plasminogen to the cell surface and thereby enhancing the migratory potential of cells.<sup>43</sup>

HRG itself is highly sensitive to cleavage by plasmin. The plasmin-cleaved HRG exhibits lower affinity for cell surface heparan sulfate but enhanced affinity for necrotic cells and plasminogen. Both intact and plasmin-cleaved HRG enhanced the binding of plasminogen to heparin-coated surfaces to a similar

extent. Furthermore, heparin, Zn<sup>2+</sup> or acidic pH was found to protect HRG from cleavage by plasmin. Thus proteolytic cleavage of HRG by plasmin may provide a feedback mechanism to regulate the effects of HRG on the plasminogen/plasmin system and perhaps other functions of HRG.<sup>44</sup>

## FUNCTIONS IN ANGIOGENESIS, ANTIBACTERIAL EFFECTS, AND IMMUNE SYSTEM

As described earlier, HRG binds to thrombospondin and inhibits its antiangiogenic properties.<sup>40</sup> However, a His-rich peptide (HRG330–364) can interact with cell surface heparan sulfate and exert its antiangiogenic effect.<sup>34</sup> Zn<sup>2+</sup>-dependent binding of the His-rich domain of HRG to heparan sulfate on endothelial cells is required for the inhibition of angiogenesis, providing a novel mechanism to increase the local concentration and activity of HRG, which may reflect a host response to counteract angiogenesis during pathological conditions. The finding that tumor angiogenesis is elevated in HRG-deficient mice supports this argument.<sup>45</sup>

The peptide (GHHPH)<sub>4</sub>, derived from the His-rich domain of HRG, containing a heparin-binding motif, was antibacterial for *E. faecalis* in the presence of Zn<sup>2+</sup> or at low pH. The results show a previously undisclosed antibacterial activity of HRG and suggest that the His-rich and heparin-binding domain of HRG is responsible for this antibacterial activity.<sup>14</sup>

HRG promotes adhesion and morphological changes of a human T-cell line (MOLT-4) in a concanavalin A-dependent manner. HRG/Con A cooperate to activate leukocytes via a signaling pathway distinct from that of Con A alone,<sup>46</sup> and cell surface ATP synthase is likely to function as a receptor for HRG on MOLT-4 cells.<sup>47</sup>

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